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Horizontal and vertical water and solute fluxes in paddy rice fields

Manon Janssen *, Bernd Lennartz

Institute for Land Use, Rostock University, Justus-von-Liebig-Weg 6, D-18059 Rostock, Germany

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Abstract

Water scarcity is an increasing problem in water-consumptive irrigated rice landscapes. This study quantified water losses in paddy rice fields due to vertical percolation through the plough pan and due to lateral water fluxes within the puddled layer. The average infiltration rates (geometric mean) for three paddy fields with a cultivation history of 3, 20 and 100 years were 28.0, 0.79 and 0.16 cm/day, respectively, demonstrating a strong dependence of the infiltration rate from the age of the field. Puddling reduced the percolation rate about 35-fold after 20 years and 175-fold after 100 years of paddy cultivation, confirming the importance of maintaining an undisturbed, permanent plough pan in order to increase water use efficiency. Lateral flux experiments revealed that horizontal preferential flow drove water and solute fluxes in the cracked topsoil of paddy fields over a flow distance of 50 cm, indicating high water losses during land preparation even with an existing plough pan. Lateral preferential transport of water and solutes towards unpuddled spots within the field or towards permanent cracks or root macropores in the plough pan is facilitated.

Keywords: Plough pan; Puddling; Percolation losses; Cracked soil; Flux heterogeneity; Preferential flow

1. Introduction

Rice is the most important staple food in Southeast China, accounting for half of the total grain output and one fourth of the arable land in China (Gong and Xu, 1990). In irrigated rice landscapes, problems related to water quality and water scarcity are increasing due to the intensification of agriculture, and increasing water consumption for household and industrial purposes. Therefore, the development of water saving strategies for water-consumptive rice cultivation in regions such as the Liu Jia Zhan district is of major concern (Xiaoping et al., 2004; Zhao, 2001). One of the important proposed measures is the restriction of vertical water percolation and lateral seepage.

Land preparation for rice cultivation starts with flooding of fields for at least two days. Afterwards, the soil is puddled at a water content between field capacity and saturation, comprising repeated ploughing, harrowing and finally levelling (Kukal and Aggarwal, 2002; Cabangon and Tuong, 2000; Wopereis et al., 1992b). Puddling assists weed control and homogenization of the soil by destroying aggregates and macropores. The low mechanical strength of the puddled layer allows easy transplanting (Ringrose-Voase et al., 2000; Wopereis et al., 1992b). As a result of the activity of animal hooves or machines, a compacted zone (plough pan) is formed underneath the puddled layer. This process is amplified by settling and consolidation of dispersed clay particles, forming a semi-permeable layer at the bottom of the puddled layer which blocks

* Corresponding author. Tel.: +49 381 4984092; fax: +49 381 4982159.
E-mail address: manon.janssen@uni-rostock.de (M. Janssen).

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pores and further reduces the saturated hydraulic conductivity of the plough pan (Kukal and Aggarwal, 2002; Tuong et al., 1994).

The effect of puddling on water percolation has been intensively studied with respect to puddling intensity and depth (Kukal and Aggarwal, 2002; Singh et al., 2001; Painuli et al., 1988), thickness of the plough pan (Wopereis et al., 1994b), soil type (Sharma and Bhagat, 1993), ponding water depth (Wopereis et al., 1994b), the effect of non-puddled spots on field percolation (Tuong et al., 1994) and seasonal variability within one cropping season (Kukal and Aggarwal, 2002; Tuong et al., 1994). All of these investigations were conducted under well-defined experimental boundary-conditions, while long-term effects of puddling at the field scale have not been studied yet.

In continuously irrigated fields that are drained once before harvest, cracks will form after drainage and remain until next season's puddling. Some cracks, however, might even persist after rewetting and puddling (Cabangon and Tuong, 2000). As a new cultivation technique, intermittent irrigation is increasingly practised to increase water use efficiency. In clayey paddy soils, this temporary flooding with drying cycles may lead to the formation of shrinkage cracks also during the irrigation period, increasing the risk of preferential water losses (Ou et al., 1999). Laterally extending cracks (in the puddled layer) are on the one hand advantageous during land preparation as they ensure a fast and even spreading of the irrigation water over the entire field, but on the other hand, they are a major reason for high water losses (Cabangon and Tuong, 2000). Tuong et al. (1994) found that during reflooding, about 70% of the bypass flow water is lost to adjacent fields and canals due to lateral drainage. Wopereis et al. (1994a) reported that preferential flow (or bypass flow) significantly contributes to total field fluxes in cracked clay soils at the onset of the monsoon season, when water starts ponding in the cracks on top of the impermeable plough pan. Cracks may also preferentially transport water laterally to non-puddled spots within the field (Tuong et al., 1996), and thereby increase vertical percolation rates.

The objectives of this study were to quantify water losses in paddy rice fields (i) due to vertical percolation through the plough pan and (ii) due to lateral water fluxes within the puddled layer, and (iii) to identify solute transport mechanisms. The influence of the cultivation duration on the impermeability of the plough pan was comparatively investigated examining fields that had been used as paddies for 3, 20 and over 100 years, respectively.

2. Materials and methods

2.1. Study sites

The field experiments were conducted in south-eastern China in the vicinity of the Ecological Experimental Station of Red Soil, Liu Jia Zhan Township, Jiangxi Province (116.90°E and 28.23° N). This region is characterized by a warm and humid subtropical monsoon climate with an annual mean air temperature of 17.8 °C and an annual rainfall of 1706 mm (1954–1999), whereof 50% falls from March to July (Zhang and Horn, 2001). Frequent drought hazards occur from July to September. The prevailing soils have a loam and clay loam texture and have developed from loose, easily eroded, quaternary red clays, which are widely distributed in the inter-montane basins of South China. The relief is slightly undulating with elevation differences of 40 m at the maximum. Two irrigated rice crops are grown from May to July and August to October with green manure during the winter period. Land cultivation is performed manually or with the help of water buffalos. All field experiments reported here were carried out shortly after rice harvest in October and November 2004.

2.2. Soil properties

Three field sites within the same small catchment representing a chronosequence of agricultural use were investigated. The oldest field (QO) has been used for paddy cultivation for more than 100 years and is situated in the valley. The two other sites at higher elevations have been converted into paddy fields more recently: one about 20 years ago (QY), and the uppermost only 3 years ago (QV). The fields cover about 1100, 170 and 150 m^2, respectively. Three horizons (hydraulic-functional horizons as introduced by Wopereis et al., 1993) were identified in each profile: puddled topsoil (Ap), plough pan or hard pan (P) and subsoil (C). The puddled horizon had a thickness of 11–17 cm. No visible distinct plough pan had developed yet on field QV, while it was about 4 cm thick on field QY and 5 cm on field QO (Fig. 1). Soil types varied between the sites due to their different land use history; they were hydragric anthrosol (fields QO, QY) and anthric cambisol (QV).

The quaternary red clay parent material has a high clay content of up to 40% (Li et al., 2005), which corresponds to the subsoil of the three fields containing between 35 and 43% clay (Table 1). In the cultivated horizon and plough pan, clay contents were less

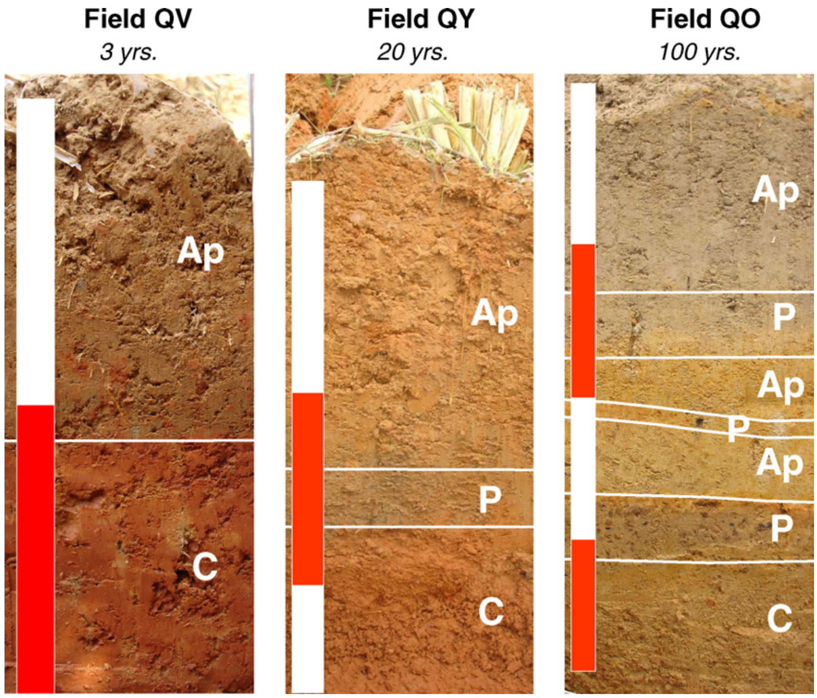


Fig. 1. Soil profiles of fields QV, QY and QO. Ap: puddled topsoil; P: plough pan or hard pan; C: subsoil.

pronounced with 37% (QV, QY) and 19% (QO) with an accordant increase in sand content. The progressive loss of clay with paddy cultivation is related to ferrolysis, to lateral runoff and to the vertical displacement of clay from the upper to the lower part of the soil profile by repeated suspension and sedimentation of clay particles during soil treatment under ponding conditions (Tuong et al., 1994; Li et al., 2005).

For laboratory investigations, undisturbed soil core samples (250 cm^3) were collected from the puddled layer with ten replicates in vertical and ten replicates in horizontal direction at each investigated field. The saturated hydraulic conductivity was determined with the constant head method.

2.3. Vertical infiltration into plough pan

The infiltration rate into the plough pan was determined using the double-ring infiltrometer method. The inner and outer ring diameters were 18 and 36 cm, respectively. The 12 (QV) and 9 (QY, QO) infiltration locations were evenly distributed over the sites. The soil of the puddled layer was removed and the rings were carefully inserted 2–3 cm deep into the plough pan.

Table 1
Soil properties of experimental sites

Field	Age (years)	Depth (cm)	Horizon	Texture class	Clay (%)	Silt (%)	Sand (%)	Bulk density (g/cm^3)
QV	3	0–12	Ap	Clay loam	37.8	39.4	22.9	1.20
		>12	C	Clay	43.1	33.5	23.4	1.49
QY	20	0–14	Ap	Clay loam	36.6	42.2	21.1	1.29
		14–18	P	Clay loam	35.5[a]	44.6[a]	20.0[a]	1.56
		>18	C	Silty clay	40.1[a]	43.0[a]	16.9[a]	1.26
QO	100	0–14	Ap	Loam	19.1	44.3	36.5	1.00
		14–19	P1	Loam	19.6	45.7	34.8	1.70
		19–29		Loam	26.0	43.9	30.2	
		29–33	P3	Clay loam	32.0	43.7	24.3	
		>33	C	Clay loam	35.1	42.1	22.8	

[a] Data from Janssen et al. (2006).

Water level in the rings was kept constant at 3 cm (field QV) and 10 cm (fields QY, QO), respectively. For field QV, where an initial high infiltration rate decreased monotonically towards a constant value with time, the steady-state infiltrability was assessed by fitting Philip's equation to the experimental data (Philip, 1957):

$$i = \frac{1}{2}St^{-1/2} + A \qquad (1)$$

where i is the infiltration rate (cm/day), S the sorptivity (cm day$^{-1/2}$), t the time (day), and A (cm/day) is a soil parameter representing the final infiltration rate. The steady-state infiltration rate was thus set to parameter A for field QV. At the two sites QY and QO, however, almost no time depending variations of the (comparably low) infiltration rate was observed during the measurement period of about 24 hours. Measured infiltration rates followed more or less a horizontal straight line when plotted against time. Therefore, no fitting of Philip's equation to the data was possible and necessary, and for each measuring location, the arithmetic mean of all values from one measurement cycle served as the steady-state infiltrability.

2.4. Experimental set-up for measurements of lateral water and solute fluxes

Lateral flux properties of the topsoil were investigated over a flow length of 50 cm (Fig. 2). At the backside boundary of the experimental plot, a ditch was dug down to the top of the plough pan. A hydraulic gradient was imposed by adjusting and maintaining (Mariott vessel) a water level of 11 cm within the pit just below the soil surface. On field QY, fluxes were too high and a water level of only 5–7 cm could be ensured.

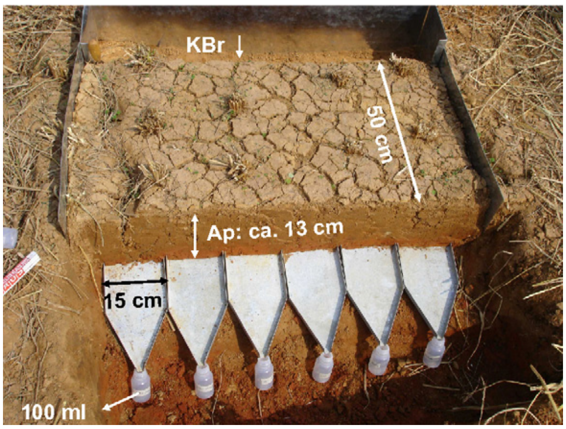


Fig. 2. Experimental set-up of lateral flux experiment.

Potassium bromide was added as a conservative tracer to the infiltrating water with a concentration of 2100 mg/L. The sidewalls of the experimental plot were sealed by metal sheets, which were driven down into the plough pan in order to prevent lateral water losses. The discharge resulting from the imposed gradient was captured spatially resolved by 15 cm wide metal sheets that had been inserted into the soil just above the plough pan. The fluxes were quantified with plastic bottles of varying volume according to flux intensity. Two experimental plots with a sum of 12 collector sheets were constructed at field QV, and four plots with a total of 24 collector sheets at field QO. The duration of the experiment varied between plots according to the resulting flux rate.

In order to compare the heterogeneity of lateral fluxes of both investigated sites, graphs as introduced by Quisenberry et al. (1994) were prepared. First, the fraction of total water collected by each individual sheet was calculated. Then, values from all collector sheets were ranked in descending order, added up and plotted as a function of cumulative cross-sectional area. Cumulative cross-sectional area here denotes the summation of the cross-sections contributing to the single collector sheets. Given that the flow is uniform, the graph results in a 1:1 relationship between fraction of total discharge and cumulative cross-sectional area. A heterogeneity index (HI) was calculated (Stagnitti et al., 1999), which is based on the cumulative probability density of the standard beta function (Eq. (2)):

$$p(x;\alpha,\zeta) = \frac{\Gamma(a+\zeta)}{\Gamma(\alpha)\Gamma(\zeta)}x^{\alpha-1}(1-x)^{\zeta-1}; \quad \text{for } \alpha \geq 0,\ \zeta \geq 0,\ 0 \leq x \leq 1 \qquad (2)$$

where Γ is the gamma function and α and ζ are free parameters. These two independent parameters are fitted to the cumulative discharge values. The heterogeneity index is then defined as a scaled ratio of standard deviation and mean of the beta function:

$$\mathrm{HI}(\alpha,\zeta) = \frac{\sqrt{3}\sigma_x}{\mu_x} = \sqrt{\frac{3\zeta}{\alpha(\alpha+\zeta+1)}} \qquad (3)$$

The degree of uniformity (DU) as proposed by Koszinksi et al. (2006) is also based on the cumulative beta function. It is calculated as the fraction of cross-sectional area conducting 90% of the discharge (as derived from the beta function, Eq. (2)) divided by 0.9. For the homogeneous case, 90% of the area would conduct 90% of the discharge and thus result in a DU of 1.

3. Results and discussion

3.1. Saturated hydraulic conductivity from core sample measurements

The average saturated hydraulic conductivities of the puddled layer of sites QV, QY and QO resulting from the laboratory measurements as expressed as the geometric mean of the measured values were 175, 42 and 76 cm/day, with standard deviations of 242, 97 and 95 cm/day, respectively. Testing the data with the Student's *t*-test revealed that conductivity of field QV was significantly higher than that of field QY at a confidence level of 99.5%, and significantly higher than the conductivity of site QO at a confidence level of 97.5%. Conductivities of fields QY and QO did not differ significantly. Thus, values of the youngest site QV were slightly higher than those of the older fields. It is assumed that the disaggregation of soil structural elements due to puddling is less advanced in the youngest field and that remaining aggregates still form a secondary pore system maintaining a higher hydraulic conductivity as compared to the other longer treated sites. As expected, no anisotropy effects could be observed in any of the puddled layers.

The relatively high hydraulic conductivities at all three sites might be explained by biopores, which were present in the biologically active soil and thus also in the soil cylinders that were taken for measurements. The pores might not have closed completely during saturation and have been continuous in the small cylinders, while they are not necessarily in the field soil. The 250 cm^3 soil cylinders were thus smaller than the representative elementary volume, an assumption which is confirmed by the in situ lateral flux experiments where distinctly lower conductivities have been observed (field QO, see below). Discontinuous macropores in the field explained high differences between in situ infiltration measurements and conductivity tests on detached cylinders encountered by Wopereis et al. (1992a).

3.2. Vertical infiltration into plough pan

The steady-state infiltrability values of the plough pan are presented in Fig. 3. The average infiltration rates (geometric mean) for the three fields with ages of 3, 20 and 100 years were approximately 28.0, 0.79 and 0.16 cm/day, respectively, corresponding to results reported elsewhere (i.a. Wopereis et al., 1992a).

The observed values differ significantly between fields and reveal a strong dependence of the infiltration rate from the time period the field is under paddy rice cultivation, which is due to an increasing bulk density of the plough pan (Table 1). The youngest investigated field site was chosen as a reference because of the not yet established plough pan. In this property the site represents fields that have been newly reclaimed for rice production. Its high steady-state infiltrability supports the soil classification that no distinct plough pan has developed yet.

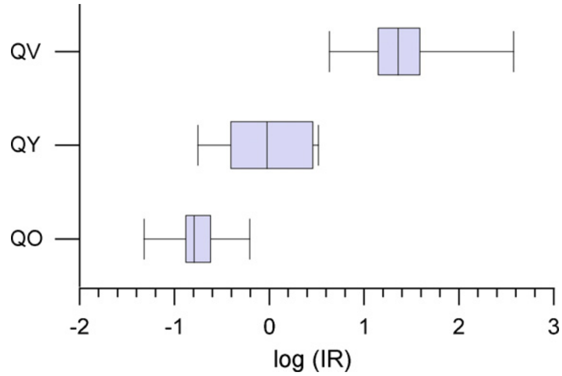


Fig. 3. Final infiltration rates (IR; cm/day) into the plough pan: boxplots of logarithmized infiltration rates.

The decrease of percolation rate as a result of puddling twice a year was about 35-fold after 20 years and 175-fold after 100 years of paddy cultivation when compared to the field that only recently has been reclaimed for cultivation. Walker and Rushton (1984) reported that the maturing process of the plough pan may take 10–20 years at most, and Liu et al. (2005) showed that infiltration rate does not decrease further after 14 cycles of ploughing and compaction. In contrast, our results show that the developing process of the plough pan may take several decades if no machines are used for land cultivation.

Comparing the hydraulic conductivities of the puddled layer and the plough pan yields ratios of 6.3, 53 and 475 for fields QV, QY and QO, respectively. Under flooded conditions, these contrasting conductivities cause the formation of a saturated puddled layer, while the major part of the plough pan as well as the subsoil remain unsaturated.

3.3. Lateral water fluxes in the topsoil

At the experimental site QO all 24 metal collector sheets delivered discharge with a lateral flux rate that varied between 1 and 25 cm/day (Fig. 4a and b). The flux slightly decreased with experimental duration for sheets that had a high initial discharge, while it slightly increased at those collector sheets that had a low rate at early stages of the experiment.

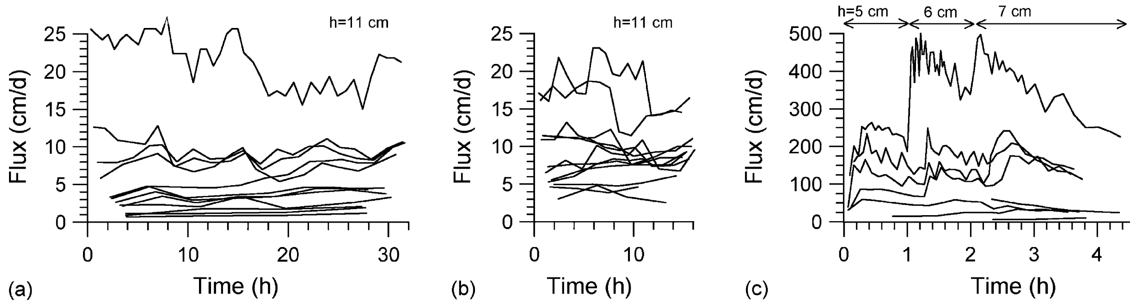


Fig. 4. Lateral fluxes per collector sheet after a flow length of 50 cm, h: adjusted water level. (a) Field QO, sheets 1–12; (b) field QO, sheets 13–24; (c) field QV, sheets 1–12.

At the youngest site (QV), the measured flux per sheet amounted up to 500 cm/day, but 4 out of 12 collector units did not yield any water (Fig. 4c). The high maximum discharge was possibly caused by a system of cracks and fissures that had developed prior to onset of the lateral infiltration tests (Fig. 5). At some collector sheets the flux rate showed no dependency to the adjusted gradient in the range of $\Delta h/\Delta x = 5/50–7/50$ cm/cm. In one case flux reduced dramatically during the test period after an initial high rate indicating either the closure of conducting cracks due to swelling or the increasing importance of vertical fluxes with increasing water content of the subsoil. Taking into consideration the saturated hydraulic conductivity as determined in the laboratory and with double-ring infiltrometers, it can be hypothesized that the observed lateral movement occurred more or less exclusively through preferential paths. In case of a homogeneous flux field, the dominant vertical fluxes through the subsoil (~28 cm/day) would have hardly resulted in any lateral water movement over a 50 cm travel distance in the puddled layer, as a 2D numerical model evaluation confirmed (results not shown here).

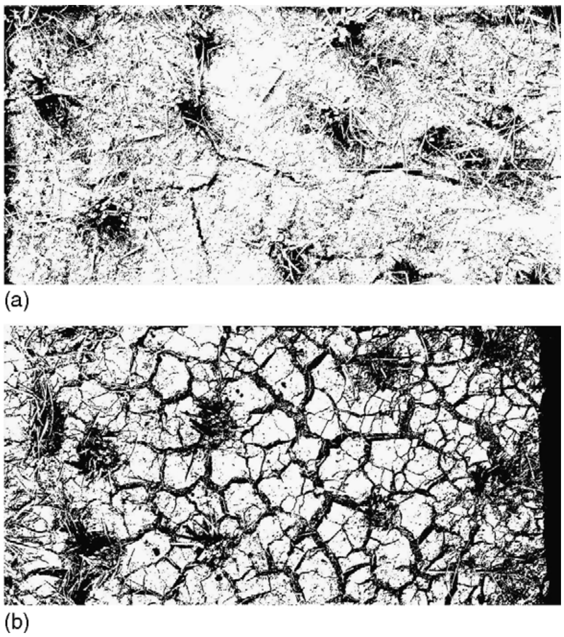


Fig. 5. Crack structure of experimental plots (50 cm × 100 cm). (a) Field QO; (b) field QV.

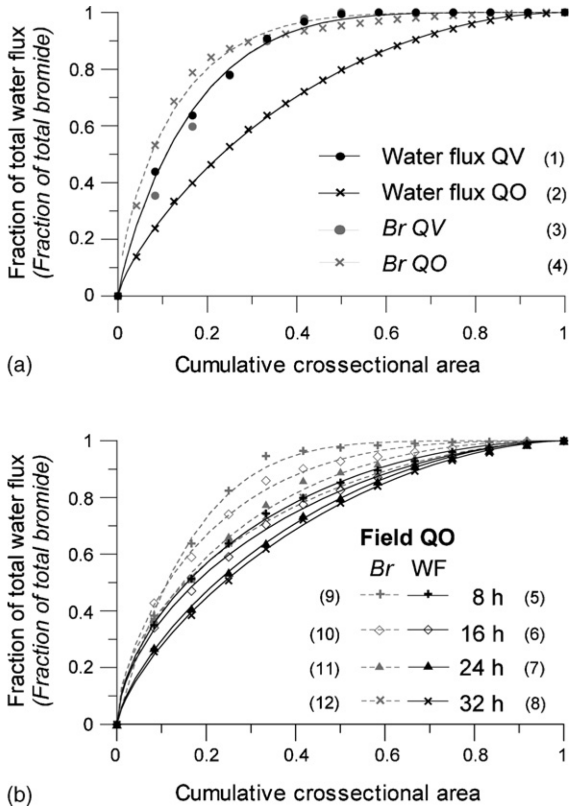


Fig. 6. Cumulative fraction of lateral water flux and fraction of total leached bromide as a function of cumulative cross-sectional area. Numbers in brackets refer to Table 2. y-Axis labels: non-italic label refers to water flux, italic label refers to bromide. (a) Comparison of fields QO and QV; (b) field QO, sheets 1–12: variation as a function of experiment duration; WF: water flux.

We plotted the cumulative flux versus flow cross-sectional area aiming at visualizing possible flow heterogeneities at both investigated sites (Fig. 6a). A homogeneous flux field would result in a 1:1 graph. From Fig. 6a it becomes evident that the flux of the youngest field (QV) is heterogeneous as compared to site QO. Ninety percent of the discharge was conducted by only 34% of the cross-sectional area on site QV, and by 65% of the area on site QO, resulting in a respective DU of 0.38 and 0.72 (Table 2). The differences of the initial water content at onset of lateral infiltration had resulted in soil structural discrepancies at both sites. While the lower soil water content at field QV enabled the formation of a crack system, the water content at site QO, which was close to saturation, kept the topsoil in a more or less non-aggregated status. The crack system at site QV promoted a preferential lateral water flux. Heterogeneity indices of 1.59 (QV) and 1.42 (QO) also confirm a more heterogeneous flux at site QV. Fig. 6b shows the development of the flux heterogeneity with experimental duration on field QO (after 8, 16, 24 and 32 hours). A slight decrease in heterogeneity with time can be stated, which is supported by DU values rising from 0.64 to 0.76, indicating that the water conducting cross-sectional area increased with continued wetting. Heterogeneity indices do also decrease from 1.69 to 1.44, but are not appropriate to describe our data in this case: HI for field QO after 8 hours (Fig. 6b, graph no. 5) is far higher than for field QV (Fig. 6a, graph no. 1), although curvature is less pronounced. Discrepancies between HI values for field QO in Figs. 6a and b (graph nos. 2 and 5) are due to a different number of collector sheets included in the calculations.

Table 2
Heterogeneity index (HI) and degree of uniformity (DU) for water and bromide fluxes

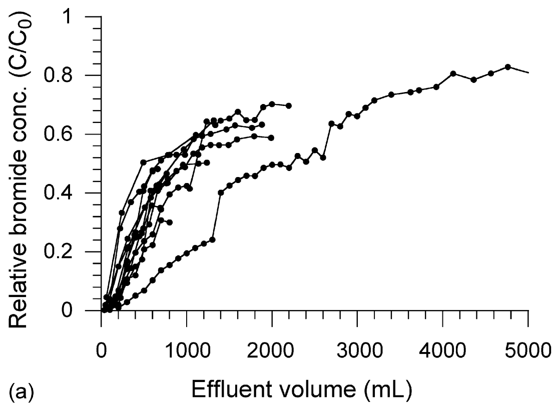
No.	HI	*A*	DU
1	1.59	0.34	0.38
2	1.42	0.65	0.72
3	1.59	0.34	0.38
4	1.83	0.29	0.32
5	1.69	0.58	0.64
6	1.65	0.63	0.70
7	1.47	0.65	0.72
8	1.44	0.68	0.76
9	1.50	0.33	0.36
10	1.76	0.43	0.48
11	1.67	0.53	0.59
12	1.75	0.61	0.68

No.: number of data set (see Fig. 6a and b); HI: heterogeneity index; *A*: fraction of area contributing 90% to total discharge/bromide concentration; DU: degree of uniformity (DU = *A*/0.9).

3.4. Lateral solute fluxes in the topsoil

Breakthrough curves of field QV show that bromide concentration in the outflow rose instantaneously after application of the tracer. For each collector sheet, the first 100 ml sample carried more than 50% of the initial concentration, confirming pronounced preferential water movement (Fig. 7b). For field QO, breakthrough curves exhibited an overall stronger variation, but only weak preferential behaviour (Fig. 7a).

A depiction of the bromide breakthrough curves against eluted pore volumes was not possible because of a variable water content within the experimental plot. However, a numerical simulation revealed that about half of the soil profile was participating in water conduction, corresponding to a pore volume of 2300 and 2050 cm^3 per collector sheet for fields QO and QV. Porosities estimated from core samples were 0.62 and 0.55 cm^3/cm^3, respectively. The observed bypass flow (Fig. 6a), however, suggests a distinctly reduced effective porosity for field QV.



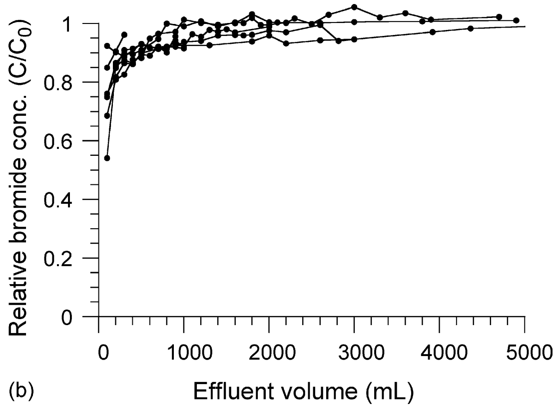


Fig. 7. Breakthrough curves of bromide tracer (see text for estimation of pore volumes). (a) Field QO, sheets 1–12; (b) field QV.

The respective heterogeneity graphs of cumulative water fluxes and total leached bromide compare to each other for field QV, with identical values of HI and DU (1.59 and 0.38, respectively) (Fig. 6a, Table 2). High water fluxes are associated with high amounts of leached bromide (correlation coefficient of 0.99), indicating that cracks, representing preferential pathways, were responsible for a fast solute transport and were continuous in the experimental plot.

For field QO, however, bromide transport was more heterogeneous than water fluxes. The HI value for bromide is 1.83 as compared to 1.42 for water fluxes, and only 29% of the cross-sectional area was responsible for 90% of the total leached bromide (Table 2). A comparison between water fluxes and total leached bromide for each collector sheet yields a correlation coefficient of only 0.76; high water fluxes were not always related to high bromide concentrations. These discrepancies can be explained by different flow mechanisms active at the same time. If the bromide solution displaces antecedent soil water instead of bypassing the soil matrix, then the bromide concentration in the outflow is low, while in other parts of the experimental plot preferential flow occurred with corresponding high bromide concentrations. This indicates that cracks were not continuous in site QO. A closer look at the four individual experimental plots of field QO confirms that correlation between water fluxes and total bromide is higher within the subplots (correlation coefficients of 0.89–0.98) and suggests different prevailing water flow paths: while preferential flow was an important feature in two subplots (collector sheets 1–12), it was not significant at the other two subplots (collector sheets 13–24). Heterogeneity of the solute fluxes decreases with experimental time, but remains superior to water flux heterogeneity throughout the experiment (Fig. 6b, Table 2). These results confirm the impact of soil structure on solute transport in paddy rice fields.

Chen et al. (2003) identified in a column infiltration experiment that horizontal nitrate transport in paddy fields was mainly controlled by concentration gradient and water potential gradient within 20 cm from the tracer source, and by matric potential beyond this distance. Our results show, however, that these driving forces may be less important under field conditions.

4. Conclusions

The results showed that infiltration rates measured on the plough pan of loamy paddy rice fields decreased 35-fold after 20 years of cultivation and even 175-fold after 100 years of cultivation, indicating that the maturing process of the plough pan may take several decades. Most field and laboratory studies so far have focussed on well-controlled short-term experiments, often with mechanical soil treatment, and revealed a much faster reduction of the vertical percolation rate. These results seem not to be valid for field situations in regions that are treated manually or with buffalos. Our results confirm the importance of maintaining an undisturbed, permanent plough pan in paddy rice fields in order to reduce percolation rate and increase irrigation efficiency. This is of special interest in areas where no winter crops (such as wheat) are grown, where a plough pan might limit plant growth and yield of the winter crop due to water logging.

Horizontal preferential flow was shown to drive water fluxes in cracked topsoils of paddy fields over a flow distance of 50 cm, indicating high water losses during land preparation even with an existing plough pan. Soil structure also governed the extent of solute transport, which was distinctly enhanced and showed preferential breakthrough behaviour in the cracked soil. The experiments were conducted with a pressure gradient often adjusted during land preparation and flood irrigation, but as well occurring naturally during and after heavy rainfall at a dry field when water starts to pond in distinct spots on the field. The results suggest that pressure heads on the plough pan may increase instantaneously in the vicinity of cracks. Moreover, the results prove that lateral preferential transport of water and solutes towards unpuddled spots within the field (as proposed by Tuong et al., 1994), towards permanent cracks or root macropores in the plough pan, but also to the higher permeable bunds may occur and thus be a threat to groundwater quality. This is particularly important in areas where intermittent irrigation is performed, because cracks form repeatedly even during the irrigation period.

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